
When Do Diffusion Transformers Outperform U-Nets?

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Abstract

1 Diffusion transformers (DiT) have recently emerged as an alternative to U-Nets
2 for denoising in diffusion models, but their reported improvements are typically
3 demonstrated at large scale, making it unclear whether the gains stem from the
4 architecture itself or from scale. We present a controlled comparison between a
5 convolutional U-Net and a patch-based diffusion transformer, trained with matched
6 parameter budgets on CIFAR-10. Beyond standard generation quality, we evaluate
7 both architectures under structured corruptions: patch masking, block occlusion,
8 and stripe removal. These corruptions degrade local cues and demand long-range
9 spatial reasoning from the denoiser. We additionally perform ablation studies over
10 transformer patch size and model depth. Our results offer insight into the conditions
11 under which global self-attention provides a genuine advantage over convolutional
12 locality for the denoising task.

13 1 Introduction

14 Diffusion models learn to generate data by reversing a gradual corruption process: given a clean
15 sample, Gaussian noise is incrementally added until the original signal is destroyed, and a neural
16 network is trained to undo each step of this corruption (Ho, Jain, and Abbeel 2020). Since their
17 introduction, diffusion models have become a dominant paradigm in image generation, producing
18 samples that rival or surpass those of generative adversarial networks while offering more stable
19 training dynamics (Dhariwal and A. Nichol 2021).

20 The architectural backbone of many early diffusion models is the U-Net (Ronneberger, Fischer,
21 and Brox 2015), a convolutional encoder-decoder with skip connections originally developed for
22 biomedical image segmentation. In diffusion models, its convolutional structure gives a natural bias
23 toward local spatial patterns, since each layer operates over a limited receptive field and broader
24 context is built up gradually through downsampling. This seems well-suited for denoising, where
25 much of the important structure is local, and U-Net-based diffusion models have achieved strong
26 results across many benchmarks (Ho, Jain, and Abbeel 2020; Rombach et al. 2022).

27 More recently, several works have explored transformer-based alternatives. Diffusion Transformer
28 (DiT) (Peebles and Xie 2023) replaces the U-Net with a patch-based transformer using self-attention
29 and reports very strong results at scale. U-ViT (Bao et al. 2023) also showed competitive performance
30 using a transformer backbone with skip connections. More recent systems such as Stable Diffusion 3
31 (Esser et al. 2024) have also moved in this direction.

32 This raises a natural question: are transformers structurally better denoisers, or do they mainly benefit
33 from scaling well? Much of the evidence so far comes from huge models trained on massive datasets,
34 making it difficult to separate architectural effects from scaling effects. In this project, we study this
35 question in a small and controlled setting, keeping the diffusion setup, optimizer, training objective,
36 and parameter count fixed while varying only the denoiser architecture. Specifically, we compare a
37 standard U-Net against a patch-based diffusion transformer trained on CIFAR-10 (Krizhevsky and
38 Hinton 2009), with the goal of isolating whether architectural inductive biases drive the performance
39 differences reported in the literature rather than scale.

40 Beyond measuring FID, we also probe whether transformers naturally benefit from self-attention that
41 models long-range dependencies directly as opposed to U-Nets that use downsampling. This factor
42 may be especially relevant when local information is absent or ambiguous.

43 To test this, we evaluate both models under structured corruptions before adding diffusion noise.
44 More specifically, we mask contiguous patches, remove large blocks, or erase alternating stripes, and
45 then test how well the models reconstruct the original image. The goal is to create settings where
46 recovering the image may require more global reasoning. If transformers benefit significantly from
47 global receptive fields, the effect should be more pronounced under these corruptions.

48 The paper focuses on three key contributions:

- 49 1. A controlled comparison of U-Net and DiT denoisers on CIFAR-10 under roughly matched
50 parameter budgets.
- 51 2. A structured-corruption evaluation setup for testing robustness under long-range reconstruc-
52 tion challenges.
- 53 3. Ablations varying transformer patch size and depth to study what aspects of the architecture
54 matter most.

55 Together, these experiments are designed to disentangle the role of architectural inductive biases from
56 the effects of scale that dominate the existing literature. If DiT outperforms under these structured
57 corruptions, that would support the idea that attention provides an architectural advantage beyond
58 scaling. Otherwise, the results may suggest the advantage seen in prior work comes mainly from
59 large-scale training.

60 2 Related Work

61 **Diffusion models.** Denoising diffusion probabilistic models were introduced by Ho, Jain, and
62 Abbeel (2020) who showed that a U-Net trained to reverse a Gaussian noising process could generate
63 high-quality images. Subsequent work improved sample quality through better noise schedules and
64 architectural refinements (A. Q. Nichol and Dhariwal 2021), and Dhariwal and A. Nichol (2021)
65 demonstrated that diffusion models could surpass GANs on class-conditional ImageNet generation
66 by using larger U-nets with additional attention layers. Latent diffusion models (Rombach et al.
67 2022) reduced the computational cost by applying the diffusion process in a compressed latent space,
68 enabling high-resolution generation with moderate compute. Throughout this line of work, the
69 convolutional U-Net remained the default denoiser architecture. Our project uses the same discrete-
70 time DDPM framework and training objective, but focuses on comparing the denoiser backbone
71 rather than improving the surrounding pipeline.

72 **Transformer-based denoisers.** Peebles and Xie (2023) proposed the Diffusion Transformer (DiT),
73 which replaces the U-Net with a vision transformer operating on *patchified* latent representations.
74 They showed that increasing transformer compute, through depth, width, or token count, consistently
75 improved FID, with larger DiT models achieving state-of-the-art results on ImageNet. Concurrently,
76 (Bao et al. 2023) introduced U-ViT, a transformer backbone with U-Net-style skip connections, and
77 showed competitive or improved performance over convolutional U-Nets on benchmarks including
78 unconditional CIFAR-10 generation. DiffiT (Hatamizadeh et al. 2024) further explored transformer-
79 based diffusion using time-dependent self-attention, also reporting strong CIFAR-10 results in pixel
80 space. More recent models such as PixArt- α (Chen et al. 2024) and Stable Diffusion 3 (Esser et al.
81 2024) also use transformer backbones in large-scale text-to-image generation.

82 Our project differs from this line of work in two ways. First, rather than studying scaling benefits, we
83 focus on whether transformers retain an advantage at small scale under roughly matched parameter
84 budgets. Second, prior work has not, to our knowledge, examined robustness under structured
85 corruptions intended to probe long-range dependencies, which is one focus of our experiments.

86 **Scaling studies.** Li et al. (2024) conducted a controlled comparison of U-Net and transformer
87 backbones for text-to-image diffusion under matched training conditions. They found transformer
88 depth to be a parameter-efficient scaling direction, while also identifying efficient U-Net variants that
89 remained competitive. However, their experiments operate at much larger scales than we consider.

90 Tian et al. (2024) proposed U-DiT, which combine token downsampling with transformer blocks,
 91 and showed strong quality-efficiency tradeoffs.

92 These works are related to ours, but ask a somewhat different question: how to scale transformer
 93 diffusion models effectively, whereas we ask whether transformers help at all in a low-budget regime.

94 **Inductive biases of CNNs and transformers.** The question of when transformers outperform
 95 convolutions has also been studied in discriminative settings. Dosovitskiy et al. (2021) showed that
 96 Vision Transformers often require large-scale pre-training to match CNN performance, suggesting
 97 weaker inductive bias can be a disadvantage in low-data settings. Raghu et al. (2021) found that ViTs
 98 and CNNs develop different representations, with transformers tending to capture global structure
 99 earlier. Naseer et al. (2021) showed transformers can be more robust to some image corruptions,
 100 possibly due to their ability to model long-range dependencies.

101 In generative settings, this question is less settled. Park and Kim (2022) studied interactions between
 102 attention and convolution and found self-attention may improve optimization in some settings. Partly
 103 motivated by this line of work, our structured corruption experiments ask: if transformers do help
 104 with long-range dependencies, does it become more visible when local structure is disrupted?

105 3 Method

106 3.1 Diffusion Framework

107 Both models are trained within a standard discrete-time DDPM framework (Ho, Jain, and Abbeel
 108 2020). Given a clean image x_0 , the forward process produces a noisy image x_t at timestep $t \in$
 109 $\{1, \dots, T\}$ according to

$$x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1 - \bar{\alpha}_t}\varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I), \quad (1)$$

110 where $\bar{\alpha}_t = \prod_{s=1}^t (1 - \beta_s)$ and $\{\beta_t\}_{t=1}^T$ is a linear noise schedule with $\beta_1 = 10^{-4}$ and $\beta_T = 0.02$.
 111 We use $T = 500$ timesteps. A denoising model $f_\theta(x_t, t)$ is trained to predict the added noise ε by
 112 minimizing the objective

$$\mathcal{L}(\theta) = \mathbb{E}_{x_0, \varepsilon, t} [\|\varepsilon - f_\theta(x_t, t)\|_2^2]. \quad (2)$$

113 This objective is shared by both architectures; the only difference between the two experiments is the
 114 denoiser f_θ itself.

115 3.2 U-Net Baseline

116 Our U-Net follows the standard DDPM design. The encoder consists of residual blocks with group
 117 normalization and SiLU activations, organized across multiple resolution levels with channel widths
 118 determined by a base channel count and a set of multipliers. Timestep information is injected by
 119 projecting a sinusoidal positional embedding through an MLP and adding it to the intermediate
 120 activations within each residual block. A multi-head self-attention block is applied at the bottleneck
 121 resolution. The decoder mirrors the encoder, with skip connections that concatenate encoder features
 122 at each resolution level.

123 We use a base channel width of 32 with multipliers (1, 2, 2), yielding channel widths of 32, 64, and
 124 64 at successively lower resolutions. The model operates at CIFAR-10’s native 32×32 resolution.

125 3.3 Diffusion Transformer

126 Our DiT follows the design of (Peebles and Xie 2023). The input image is split into non-overlapping
 127 $p \times p$ patches, each of which is linearly projected to produce a token embedding. Learnable position
 128 embeddings are added, and the resulting sequence is processed by a stack of transformer blocks. Each
 129 block consists of multi-head self-attention and an MLP with GeLU activation. Timestep conditioning
 130 is injected via adaptive layer normalization with zero initialization (adaLN-Zero): the timestep
 131 embedding modulates the scale, shift, and gating parameters of each sub-layer, with all modulation
 132 parameters initialized to zero so that each block begins as the identity function. The output tokens are
 133 linearly projected back to pixel space and rearranged into the original image layout.

134 We use a patch size of 4 (yielding $8 \times 8 = 64$ tokens for a 32×32 image), a hidden dimension of
 135 128, 4 transformer blocks, and 4 attention heads.

3.4 Training Protocol

Both models are trained on the CIFAR-10 training set (50,000 images) with identical hyperparameters: AdamW optimizer with learning rate 2×10^{-4} , batch size 128, gradient clipping at norm 1.0, and an exponential moving average (EMA) of model weights with decay 0.9999. Training runs for 40 epochs. All reported results use the EMA model. The only data augmentation is random horizontal flipping. The U-Net has 1.30M trainable parameters and the DiT has 1.34M, ensuring a fair comparison at matched capacity.

3.5 Structured Corruption Evaluation

To test whether the transformer benefits more from global context, we evaluate both trained denoisers under a structured-corruption protocol. Given a clean test image x_0 , we:

1. Divide the image into a grid of $p_m \times p_m$ non-overlapping patches and randomly mask a fraction r of them by setting their pixel values to zero, producing a corrupted image $\tilde{x}_0$.
2. Add diffusion noise at a fixed evaluation timestep $t_{\text{eval}} = T/2$ to obtain $x_t = \sqrt{\bar{\alpha}_{t_{\text{eval}}}} \tilde{x}_0 + \sqrt{1 - \bar{\alpha}_{t_{\text{eval}}}} \varepsilon$.
3. Apply the trained denoiser to predict the noise: $\hat{\varepsilon} = f_{\theta}(x_t, t_{\text{eval}})$.
4. Reconstruct an estimate of the clean image: $\hat{x}_0 = (x_t - \sqrt{1 - \bar{\alpha}_{t_{\text{eval}}}} \hat{\varepsilon}) / \sqrt{\bar{\alpha}_{t_{\text{eval}}}}$.
5. Measure reconstruction MSE separately in the *masked* and *unmasked* regions relative to the original clean image x_0 .

In our primary experiments, we use $p_m = 8$ patches and $r = 0.5$ (i.e., half of the 8×8 patches are masked), and average results over multiple test batches. The key quantity of interest is the MSE in the masked region: this measures how well the denoiser recovers content that was absent from the corrupted input. This requires integrating information from distant visible regions.

3.6 Ablation Studies

To understand which aspects of the transformer architecture matter most, we perform ablations on two quantities: (i) Patch size (2, 4, 8): controls the number of tokens and, consequently, the granularity of global attention. Smaller patches yield more tokens and finer spatial resolution but higher computational cost. (ii) Model depth (4, 6, 8 transformer blocks): controls the network’s capacity for composing local and global features across layers.

For each ablation, we retrain the DiT variant under identical conditions and evaluate on both the standard denoising objective and the structured-corruption protocol.

4 Experiments and Results

4.1 Standard Denoising Comparison

Figure 1 shows the training loss curves over 40 epochs. The U-Net converges faster and reaches a substantially lower final loss (≈ 0.047) than the DiT (≈ 0.078), a relative gap of roughly 40%. The DiT begins at a much higher loss, likely due to the adaLN-Zero initialization, which causes each block to start as the identity, but drops rapidly within the first few epochs before plateauing. Despite this, the gap persists throughout training, suggesting that at this scale the convolutional inductive biases (locality, translation equivariance, multi-resolution processing) provide a genuine advantage that the transformer cannot overcome through training alone.

This difference is also reflected in sample quality. Using 2048 generated samples for evaluation, the U-Net obtains an FID of 247.45, while the DiT obtains an FID of 340.18. Although both FID scores are high compared to fully trained diffusion models, which is expected given our small models and limited training budget, this comparison is still informative: under matched conditions, the U-Net produces substantially better samples than the DiT. The U-Net’s generated samples (see Section A) show recognizable object structure, while the DiT samples are visibly noisier, consistent with the higher training loss.

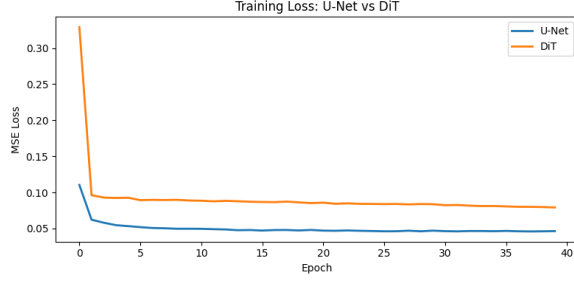


Figure 1: Training MSE loss over 40 epochs on CIFAR-10. The U-Net converges faster and to a lower final loss than the DiT under matched training conditions (1.30M vs. 1.34M parameters).

Table 1: Standard sample quality comparison using FID computed from 2048 generated samples. Lower is better.

Model	FID ↓
U-Net	247.45
DiT	340.18

182 4.2 Structured Corruption Results

183 Table 2 presents the main structured-corruption results. Contrary to our hypothesis, the U-Net
 184 outperforms the DiT in *both* the masked and unmasked regions, and by a substantial margin. The
 185 U-Net achieves a masked-region MSE of 0.754 compared to the DiT’s 1.462 (nearly twice as large)
 186 and the gap persists in the unmasked region (0.260 vs. 0.818).

187 This result suggests that, at small scale, the U-Net’s advantage is standard denoising transfers directly
 188 to the corruption settings: a model that is simply a better denoiser overall will also be better at
 189 recovering masked content, regardless of whether that recovery requires local or global reasoning.
 190 Figure 2 makes the contrast apparent: the U-Net’s reconstructions, while noisy, preserve recognizable
 191 spatial structure and approximate color in the masked regions, whereas the DiT produces nearly
 192 incoherent outputs dominated by patch-level block artifacts, reflecting the same 4×4 patch grid
 193 visible in its generated samples (see Section A). An important caveat is that single-step denoising
 194 at a fixed noise level makes it difficult to disentangle whether the DiT’s poor corruption reflects a
 195 fundamental inability to integrate global context or simply its weaker overall denoising capacity at
 196 this scale.

197 5 Discussion

198 Our main finding is that, at small scale, the convolutional U-Net provides a clear advantage over the
 199 diffusion transformer for both standard denoising and reconstruction under structured corruption.
 200 The U-Net achieves a lower training loss, better FID, and substantially lower reconstruction error in
 201 both masked and unmasked regions. This is a negative result with respect to our initial hypothesis as
 202 we expected the DiT’s global self-attention to provide a measurable advantage when the denoising
 203 problem requires long-range context integration, but it is a meaningful one.

204 **Why the U-Net wins at small scale.** The convolutional architecture encodes strong inductive
 205 biases such as locality, translation equivariance, and multi-resolution feature extraction, that are
 206 well-matched to natural image structure. At small model sizes and limited training, these biases
 207 effectively regularize the model, allowing it to learn useful features quickly. The transformer, lacking
 208 these biases, must learn spatial structure entirely from data, which may require more capacity and
 209 training time than our regime provides. Additionally, the U-Net’s encoder-decoder design processes
 210 information at multiple spatial scales simultaneously, while our isotropic DiT operates at a single
 211 resolution. The success of U-ViT (Bao et al. 2023) and U-DiTs (Tian et al. 2024), which add skip
 212 connections and multi-resolution processing to transformer backbones, suggests that these structural
 213 elements may matter more than the choice of *convolutions* versus *attention*.

Table 2: Reconstruction MSE under structured patch corruption (8×8 patches, 50% mask ratio, noise at $t = T/2$). Lower is better. The U-Net outperforms the DiT in both masked and unmasked regions.

Model	Masked Region MSE	Unmasked Region MSE
U-Net	0.754	0.260
DiT	1.462	0.818

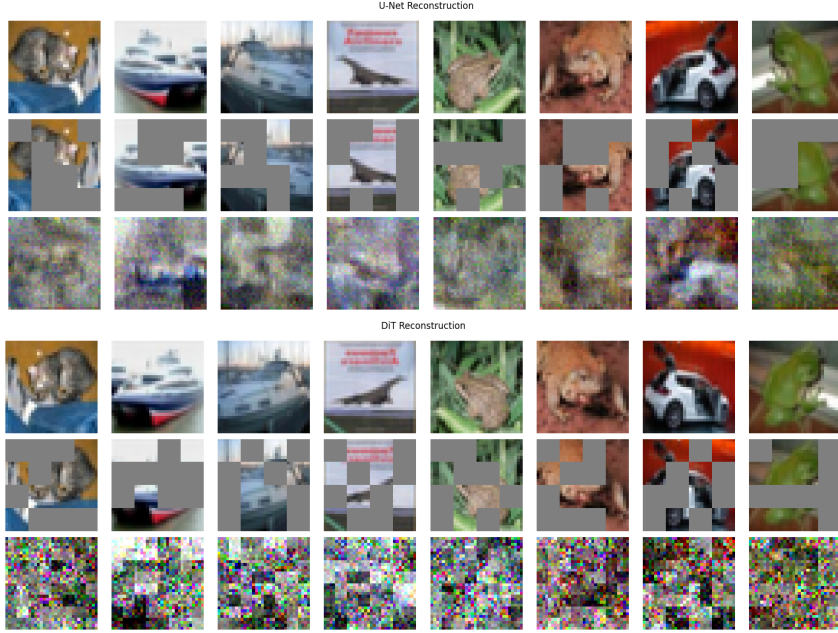


Figure 2: Reconstruction under structured corruption (50% of 8×8 patches masked, noise at $t = T/2$). Top panel: U-Net reconstructions preserve recognizable structure despite noise. Bottom panel: DiT outputs are nearly incoherent, with visible 4×4 patch-grid artifacts reflecting the model’s failure to learn meaningful spatial structure at this scale.

The role of scale. Our results are consistent with the broader pattern in the literature: DiTs achieve their strongest results at large scale (Peebles and Xie 2023), and their advantages may depend on having sufficient data and compute to overcome the lack of convolutional inductive biases. The depth ablation supports this interpretation only partially. Increasing depth from 4 to 6 improves the DiT slightly, but increasing to 8 blocks yields almost no further benefit. Thus, simply adding a few more transformer layers is not enough in this regime. The recent success of DiTs in state-of-the-art generative models may therefore reflect a favorable scaling regime together with architectural refinements, rather than an inherent advantage of plain patch-based transformers for small-scale pixel-space denoising.

Limitations. The models are considerably small (1.3M parameters) and trained for a limited duration (40 epochs), which may disadvantage the transformer disproportionately. The FID scores are therefore not meant to represent competitive CIFAR-10 generation quality; rather, they provide a relative comparison under our constrained training setup. The structured-corruption evaluation uses single-step denoising rather than a full reverse chain, conflating overall denoising quality with global context integration ability.

Future work. The most valuable extensions could be: (1) scaling both models to identify the “crossover point” where the DiT matches the U-Net, (2) repeating the corruption evaluation with a full sampling chain, and (3) testing hybrid architectures such as U-ViT (Bao et al. 2023) to disentangle the contributions of attention, multi-resolution processing, and skip connections.

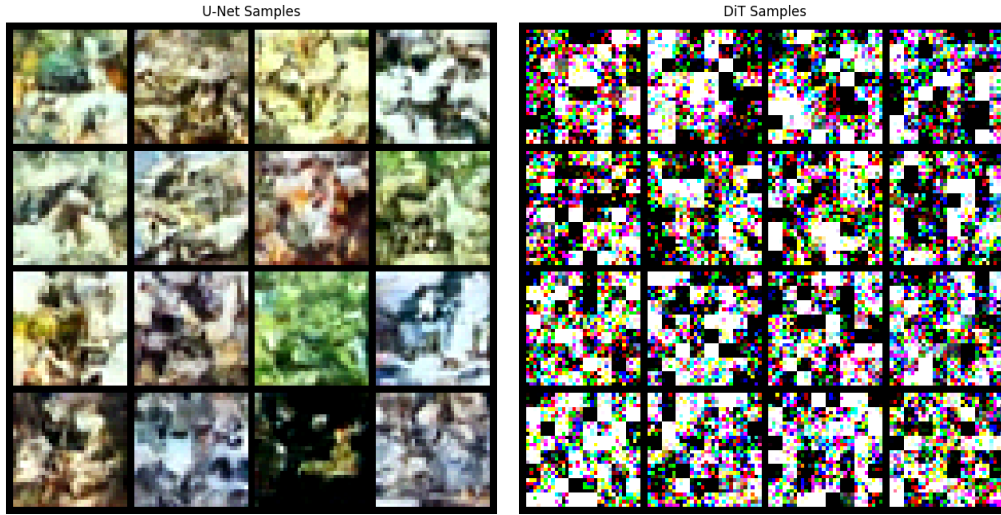
Acknowledgments

Thanks to all our day-1s.

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284 A Generated Samples



285

286 Representative samples generated by the EMA models after 40 epochs of training. The U-Net
 287 produces recognizable CIFAR-10-like images with coherent structure, while the DiT samples exhibit
 288 more noise and less spatial coherence, consistent with its higher training loss.

289 B Ablation Studies

290 **Patch size.** Table 3 shows DiT performance across patch sizes 2, 4, and 8. The default $p = 4$
 291 (64 tokens) performs best across all metrics. Reducing to $p = 2$ (256 tokens) worsens performance
 292 (despite finer resolution), likely because the quadratic attention cost over 256 tokens makes training
 293 harder within the same budget. Increasing to $p = 8$ (16 tokens) collapses performance: the final loss
 294 of 0.353 is close to untrained, indicating that 16 tokens provide insufficient spatial resolution for
 295 denoising 32×32 images.

Table 3: Patch-size ablation for the DiT. All variants trained for 50 epochs with depth 4 and hidden dimension 128. Patch size 4 provides the best trade-off between token count and spatial resolution.

Patch Size	Tokens	Final Loss	Masked MSE	Unmasked MSE
2	256	0.160	1.776	1.110
4	64	0.079	1.182	0.566
8	16	0.353	3.657	2.901

296 **Depth.** Table 4 shows the effect of increasing the DiT depth from 4 to 6 and 8 transformer
 297 blocks. Increasing depth improves performance relative to depth-4 baseline: the final loss decreases
 298 from 0.0791 to 0.0779, and the masked-region MSE decreases from 1.205 to 1.167. However, the
 299 improvement saturates quickly. Moving from depth 6 to depth 8 produces almost no additional gain.

300 This suggests that, in our small-scale setting, additional transformer depth helps up to a point but
 301 does not close the gap with the U-Net. The bottleneck is therefore unlikely to be depth alone. Instead,
 302 the DiT may need either substantially more scale, longer training, or architectural changes such
 303 as multi-resolution processing and skip connections to become competitive with the convolutional
 304 baseline.

Table 4: Depth ablation for the DiT. All variants use patch size 4 and hidden dimension 128. Increasing depth from 4 to 6 improves performance slightly, but gains saturate by depth 8.

Depth	Final Loss	Masked MSE	Unmasked MSE
4	0.0791	1.205	0.570
6	0.0779	1.167	0.545
8	0.0778	1.168	0.548

C Implementation Details

All code is implemented in PyTorch. Both models are trained on a single GPU using mixed-precision training.